

Properties of Numbers:

Property Name	What it mean	Addition example	Multiplication Example
Commutative	When you can change the order of numbers and get the same result	$a+b=b+a \rightarrow 3+5=5+3$	$a \times b=b \times a \rightarrow 4 \times 6=6 \times 4$
Associative	You can change the grouping of numbers.	$(a+b)+c=a+(b+c) \rightarrow (2+3)+4=2+(3+4)$	$(a \times b) \times c=a \times (b \times c) \rightarrow (2 \times 3) \times 5=2 \times (3 \times 5)$
Distributive	Multiply a number with a sum inside brackets.	Not Used	$a \times (b+c)=a \times b+a \times c \rightarrow 2 \times (3+4)=2 \times 3+2 \times 4$
Identity	Adding 0 or multiplying by 1 keeps the number the same.	$a+0=a \rightarrow 7+0=7$	$a \times 1=a \rightarrow 9 \times 1=9$

Summary:

- Commutative: Order doesn't matter
- Associative: Grouping doesn't matter
- Distributive: Multiply outside with each inside
- Identity: Keep number same using 0 (add) or 1 (multiply)

Write the Property Name

Expression	Property
a) $45 + 0 = 45$	
b) $56 + 97 = 97 + 56$	
c) $96 \times 40 = 40 \times 96$	
d) $5 + (6+7) = (6+5) + 7$	
e) $99 \times 1 = 99$	
f) $2 \times (5 \times 7) = (7 \times 2) \times 5$	

